

Stock Market Forecasting Without Neural Networks

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Colab Notebook

April 11, 2025

Abstract

We present a regime-based approach to stock market forecasting that bypasses the need for neural networks. Using polynomial ridge regression within a dynamic sliding window, we segment long-term historical S&P 500 data into interpretable trend regimes. From each segment, we extract statistically engineered features such as slope, R^2 , and volatility. These features are used to classify trend types (bull, bear, sideways) and generate future market scenarios via Monte Carlo simulation. Our model achieves high predictive performance ($R^2 \approx 0.98$, $\text{MAPE} \approx 12\%$), while offering greater interpretability and robustness than typical black-box models.

1 Introduction

Forecasting financial markets is inherently challenging due to their noisy and non-stationary nature. While neural networks dominate recent efforts, we argue that interpretable, feature-based classical machine learning models can be competitive and more insightful. We introduce a regime-aware forecasting framework based on polynomial ridge regression and trend feature engineering.

2 Data and Preprocessing

We use semi-annual (half-yearly) data of the S&P 500 spanning 20–25 years. Dates are parsed to timestamps, and each point is assigned a numeric time index. Using a sliding window of fixed size (e.g., 10 half-year intervals), we prepare data for polynomial regression.

3 Methodology

3.1 Polynomial Ridge Regression

For each window, we fit a quadratic model using ridge regression. Sample weights are applied to emphasize more recent data points. We then predict the next value in the sequence.

3.2 Feature Extraction

From each segment, we compute:

- **Slope**: derived from the derivative of the quadratic model.
- **R²**: coefficient of determination on the segment.
- **Volatility**: standard deviation of values within the window.

3.3 Trend Classification

We classify segments as:

- **Bull**: slope > 0.5
- **Bear**: slope < -0.5
- **Sideways**: otherwise

3.4 Clustering

Segments are grouped via KMeans clustering using extracted features. This unsupervised grouping reveals latent regime types.

3.5 Monte Carlo Simulation

Future paths are simulated by sampling historical segment statistics. For each step, slope and volatility are sampled from a random segment to update the price, enabling multiple plausible futures.

4 Results

4.1 Prediction Performance

- R²: ≈ 0.98
- RMSE: ~ 220
- MAE: ~ 170
- MAPE: $\sim 12\%$

4.2 Trend and Cluster Analysis

Boxplots and distribution charts show that prediction errors vary by trend type and cluster. Sideways trends tend to have higher uncertainty. Rolling R² confirms temporal stability.

4.3 Simulated Futures

Monte Carlo simulation provides a realistic range of future scenarios, showing growth trajectories and plausible corrections.

5 Conclusion

Our model offers an interpretable, accurate alternative to black-box forecasting. By leveraging engineered features and regime awareness, it achieves strong performance while maintaining transparency. Future work may include log-return modeling, robust regression, and better clustering.

Links

- **Colab Notebook:** https://colab.research.google.com/drive/1HcnBMpz1M0rjNG8FvDb_Cv4guoV7dSlh?usp=sharing
- **ResearchGate:** https://www.researchgate.net/profile/Pearl-Bipin?ev=hdr_xprf